

Solution Requirements

Date	15 July 2026
Team ID	XXXXXX
Project Name	AI Debt Settlement System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	The system provides secure user registration and login using JWT authentication. Only authenticated users can access loan management and AI services.	High
2	Authorization Levels	Each user can access only their own loan records, dashboard, settlement reports, and profile information. Future versions may include Admin and User roles.	High
3	External Interfaces	The system integrates with Google Gemini AI to analyze loan information and generate intelligent debt settlement recommendations.	High

4	Transaction Processing	The system allows users to add, edit, delete, and retrieve loan details. It calculates financial metrics, predicts debt risk, and generates AI-based settlement reports.	High
5	Reporting	The system generates downloadable PDF settlement reports containing borrower details, financial health analysis, debt stress level, negotiation strategy, and settlement recommendations.	High
6	Business Rules	Users must provide borrower information, loan amount, monthly income, EMI, and overdue months. The AI analyzes this data to determine financial health and recommend settlement strategies.	High
7	Compliance to Laws or Regulations	The application stores user data securely, protects user privacy using JWT authentication, and follows general data protection principles for financial information.	Medium
8	Other	The system maintains loan history, user profiles, dashboard analytics, AI-generated reports, and search functionality for efficient loan management.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	Requirement Category	Requirement Description	Priority
1	Performance	The system should generate dashboard statistics and AI settlement reports within a few seconds under normal usage.	High
2	Security	User passwords must be encrypted using bcrypt, and all protected APIs must require JWT authentication.	High

3	Usability	The application should provide a simple, responsive, and user-friendly interface that works across desktops and laptops.	High
4	Reliability	The system should maintain accurate loan records and provide consistent AI recommendations without data loss.	High
5	Scalability	The application should support increasing numbers of users and loan records with minimal performance degradation.	Medium
6	Maintainability	The project should follow a modular architecture (React Frontend, FastAPI Backend, SQLite Database) to simplify maintenance and future enhancements.	Medium